Supply Planning Decision Framework

Nabu Casa business case presentation
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Current WOS	Backlog Risk
4.8	**Critical**

New Buy	MOQ
5,000	**5,000**

A concise, discussion-led presentation based on my Streamlit decision-support prototype.

What I would do first

On Hand	Immediate Ship	Delayed PO	Backlog	New Buy
8,000	4,000	6,000	9,500	5,000

Recommendation

- Treat this as a controlled supply disruption, not just a late shipment.
- Validate recovery with the contract manufacturer before changing customer commitments.
- Use a repeatable decision sequence: optimise existing supply first, then buy only for the remaining gap.
- Publish a daily Executive Decision Report until backlog risk and supplier recovery are stable.

My planning decision engine

The framework is adapted from a planning and reporting concept I previously implemented at Apple. I used the same operating logic here because the case asks for judgement under constrained supply.

Inventory Health	**Stock Balance**	**Pull Logic**	**Push Logic**	**PO Shortage**	**Allocation**	**Backlog Risk**

Why this sequence

- Avoid unnecessary buying
- Make trade-offs visible
- Create repeatable decision ownership

What the tool adds

- Fast scenario view
- Consistent KPI logic
- A single place to explain recommendations

How I interpreted the case

Area	Assumption	Why it matters
Lead time	8 weeks manufacturing + 2 weeks shipping	Planning must use availability date, not production finish.
Open PO	10,000 includes 4,000 in transit + 6,000 delayed	Avoids double-counting supply.
Demand	Regional demand is confirmed open demand	Allocation should protect commitments.
Network	Central receipt before regional allocation	Keeps focus on planning and prioritisation.
Scope	Finished goods only	Excludes service parts, RMA and reverse logistics.

First 48 hours: control the disruption

Stabilise

- Validate shortage root cause
- Confirm 4,000 ship date
- Lock recovery date for 6,000
- Check expedite options

Decide

- Review WOS and backlog
- Assess customer commitments
- Run stock balance / pull / push
- Calculate planning deficit

Communicate

- Align Sales, Finance, Logistics
- Brief distributor-facing teams
- Share dates and caveats
- Move to daily recovery cadence

Supplier management and risk monitoring

Question	What I would confirm
Root cause	Component constraint, affected build quantity, fix owner, and recurrence risk.
Recovery plan	Committed ship dates, capacity plan, expedite options, and supplier constraints.
Customer impact	Open demand, backlog, launch risk, and order priority by region.
Controls	Daily supplier update, PO health dashboard, exception alerts, and escalation path.

Risks to monitor

- Recovery date slippage
- Backlog growth
- Distributor confidence
- Working capital pressure
- Expedite cost and logistics capacity

Allocation method for the first 4,000 units

Available	Open Demand	Gap	Priority
4,000	**9,500**	**5,500**	**US launch**

Allocation principles

- Start with confirmed customer commitments, not forecast alone.
- Protect the US launch because it has immediate commercial exposure.
- Preserve EU baseline service where possible.
- Keep APAC visible as a strategic market, even if supply is constrained.

Trade-off

- A strict allocation engine gives a transparent answer, but business judgement still matters.
- I would agree the final split with Sales and leadership before distributor communication.
- Any region receiving less than requested gets a date, rationale, and recovery cadence.

Allocation recommendation and monitoring

Region	Demand	Priority rationale	Communication message
US	4,500	Promotional launch next week	Prioritise launch coverage; confirm remaining recovery timing.
EU	3,000	Stable baseline demand	Protect key commitments; explain temporary constraint.
APAC	2,000	Strategic growth market	Keep allocation visible; avoid silent deprioritisation.

Success measures

- Fill rate by region
- Backlog aging
- Launch fulfilment
- Distributor escalations
- Recovery date adherence

Replenishment decision

Current WOS	Pipeline WOS	Target WOS	Planning Deficit	Recommended PO
4.8	10.8	12	2,076	5,000

Decision

- Place a new PO only after stock balance, pull and push options are reviewed.
- The remaining planning deficit is 2,076 units.
- Because MOQ is 5,000 units, recommended new buy is 5,000 units.
- Position this as risk protection, not speculative over-buying.

Finance justification

- Show WOS, open PO and backlog in one view.
- Explain MOQ constraint and recovery timing.
- Separate confirmed demand risk from optional buffer.
- Review weekly as forecast and supplier recovery change.

Reporting and systems I would build

Dashboard / Workflow	Purpose	Frequency
Inventory Health	WOS, inventory position, stock-out risk	Daily
Purchase Order Health	Open PO, in transit, delayed supply, revised dates	Daily
Supplier Health	Lead time, OTD, recovery commitments	Daily
Forecast Accuracy	Demand signal quality and bias	Weekly
Executive Decision Report	Single view of risk, action and owner	Daily during disruption

ERP note: the same control logic can sit in NetSuite workflows or another ERP/reporting layer.

SOP and escalation model

Supplier issue

Validate

Inventory health

Optimise supply

New buy / allocation

Daily report

Trigger	Action
Current WOS >= target	Routine monitoring
Current WOS < 8 weeks	Planner review
Current WOS < 6 weeks	Management review with Procurement
Current WOS < 4 weeks	Executive recovery meeting with supplier

Purpose: make escalation based on measurable risk, not late-stage surprise.

Governance, stakeholders and success metrics

Decision	Owner
Inventory health / stock balance	Supply Planner
Pull / push logic	Supply Planner + Procurement
New buy approval	Procurement + Finance
Allocation decision	Supply Planning Manager + Sales
Executive escalation	Operations leadership

KPIs

- Fill rate >98%
- Target WOS 12 weeks
- Supplier OTD >95%
- Backlog <1,000
- PO expedites <5%

Broader supply chain risk reduction

Supplier diversification

Qualify backup manufacturing options.

Safety stock strategy

Differentiate by demand variability and criticality.

Lead-time management

Track trends and trigger early warnings.

Concentration risk

Reduce dependency on one site where practical.

Tariffs and duties

Include landed cost in sourcing decisions.

Inventory positioning

Place stock closer to demand where justified.

What I would leave behind

A scalable decision-support rhythm

- Resolve the immediate disruption through a daily supplier recovery and allocation cadence.
- Use the decision hierarchy to justify buying decisions and avoid reactive procurement.
- Turn the Streamlit prototype logic into an operating dashboard for planners, procurement and leadership.
- Keep the discussion centred on assumptions, trade-offs and clear ownership.

How I used AI

- I had the planning framework and concept before using AI.
- The core logic was my own and based on a concept I had implemented previously at Apple.
- I used AI to help build the Streamlit prototype and improve the presentation quality.
- I reviewed and adjusted the output so the final submission reflected my own planning judgement.